2026

AI-Powered Resume Screening System with Tracing

DATA SCIENCE INTERNSHIP – FEBRUARY 2026

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1. Introduction & Objective

The objective of this project is to develop a production-ready automated resume screening system. By leveraging Large Language Models (LLMs) and the LangChain framework, the system analyses candidate resumes against a specific job description (JD). The system provides structured JSON outputs containing extracted skills, a numerical fit score, and a logical explanation for the evaluation.

2. System Architecture

The project follows a modular design pattern to separate concerns and ensure high code quality.

- ➢ Directory Structure:
  - $prompts/$: Contains specialized PromptTemplate objects.
  - $chains/$: Houses the LangChain Expression Language (LCEL) logic.
  - $main.py$: The execution script for processing multiple candidates.
- ➢ Pipeline Flow:
  - The system utilizes a linear LCEL pipeline:
    - **Input Data → Prompt Template → LLM (Gemini-2.5-Flash) → JSON Output Parser**.

3. Implementation Details

- **Prompt Engineering:** The prompt was engineered with strict negative constraints (e.g., "Do NOT assume skills") to maintain accuracy and prevent hallucinations.
- **LLM Integration:** I utilized ChatGoogleGenerativeAI to handle complex extraction tasks efficiently.
- **Observability:** Integrated **LangSmith** to monitor every step of the pipeline, providing transparency into how the AI reaches its conclusions.

4. Results & Logic Analysis

I tested the system using three distinct candidate profiles to verify the scoring accuracy:

Candidate Category	Fit Score	Reasoning
Strong	100/100	Full match for core requirements (Python, NLP, LangChain).
Average	25-40/100	Knowledge of Python but lacks domain-specific GenAI skills.
Weak	0/100	Background in an unrelated field; no technical overlap.

```
C:\Windows\System32\cmd.e

(venv) D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system>python main.py

====================
 EVALUATING: Strong Candidate
====================

{
    "extracted_skills": [
        "Python",
        "NLP",
        "LangChain"
    ],
    "experience_summary": "Senior Data Scientist with 5 years of experience in Python, and expertise in LangChain and NLP.",
    "fit_score": 95,
    "explanation": "The resume is an excellent match for the job description. It explicitly mentions all three core requirements: Python (with 5 years of experience), NLP, and LangChain. The candidate also se
lf-identifies as a 'Senior Data Scientist' and an 'Expert' in LangChain and NLP, which strongly aligns with the 'experience' requirement of the role."
}


====================
 EVALUATING: Average Candidate
====================

{
    "extracted_skills": [
        "Python",
        "Java"
    ],
    "experience_summary": "The candidate is a Software Developer proficient in Python and Java, with a stated interest in AI.",
    "fit_score": 25,
    "explanation": "The candidate explicitly matches only one of the key technical requirements: Python proficiency. The job description specifically asks for a 'Data Scientist' with experience in 'NLP' and '
LangChain'. The resume presents the candidate as a 'Software Developer' and does not mention any experience or skills related to Data Science, NLP, or LangChain. While the interest in AI is noted, it does not
 substitute for explicit experience or skills in the required domains. The significant gaps in core data science, NLP, and LangChain experience result in a low fit score."
}


====================
 EVALUATING: Weak Candidate
====================

{
    "extracted_skills": [
        "Adobe Creative Suite"
    ],
    "experience_summary": "The candidate has 2 years of experience as a Graphic Designer, utilizing Adobe Creative Suite.",
    "fit_score": 0,
    "explanation": "The resume presents a candidate with 2 years of experience as a Graphic Designer, proficient in Adobe Creative Suite. The job description, however, is for a Data Scientist requiring experi
ence in Python, NLP, and LangChain. There is no overlap in job title, domain experience, or specific technical skills between the resume and the job description. The candidate does not possess any of the expl
icitly required skills for the Data Scientist role."
}
```

Figure 1: Successful execution results showing formatted JSON for three different candidates.

5. LangSmith Tracing & Debugging

Observability is a core requirement of this project. Every execution was traced via the LangSmith dashboard.

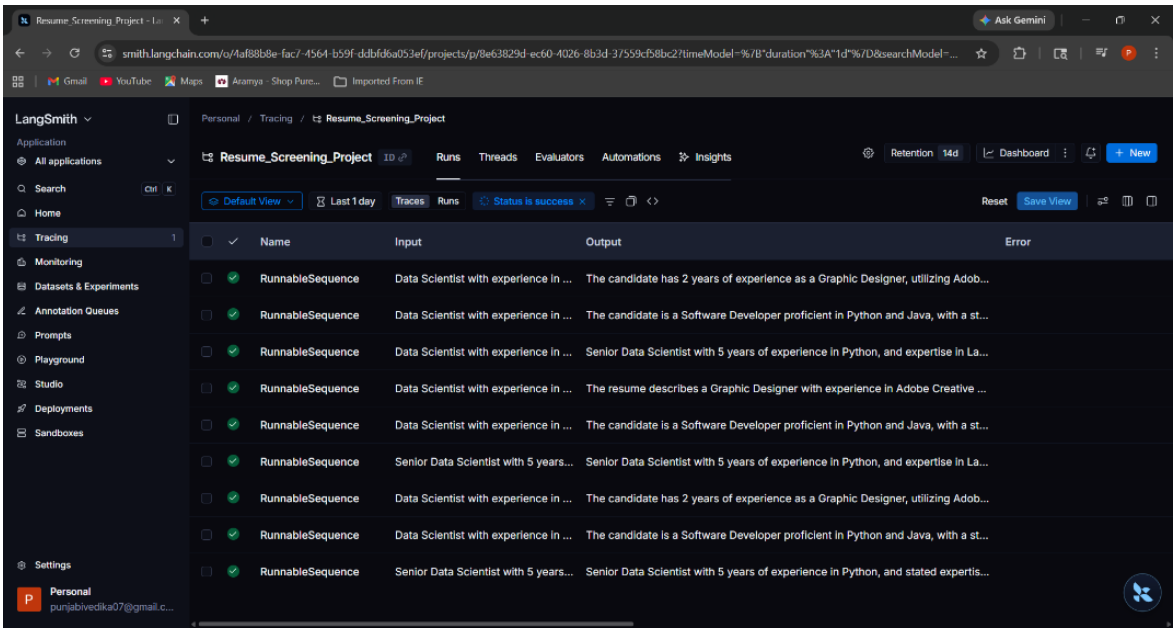

Figure 2: LangSmith project overview confirming the evaluation of three candidate profiles.

➢ Debugging Instance:

- During the development phase, I encountered a 404 NOT_FOUND error. By analysing the LangSmith trace, I identified that the API request was targeting an incorrect model version. I resolved this by explicitly defining the stable model's name in the ChatGoogleGenerativeAI configuration.

```
C:\Windows\System32\cmd.e

    ...<6 lines>...
        **kwargs,
        ^^^^^^^^^
    ).generations[0][0],
    ^
  File "D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system\venv\Lib\site-packages\langchain_core\language_models\chat_models.py", line 1185, in generate_prompt
    return self.generate(prompt_messages, stop=stop, callbacks=callbacks, **kwargs)
           ~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system\venv\Lib\site-packages\langchain_core\language_models\chat_models.py", line 995, in generate
    self._generate_with_cache(
    ~~~~~~~~~~~~~~~~~~~~~~~~~^
        m,
        ^^
    ...<2 lines>...
        **kwargs,
        ^^^^^^^^^
    )
    ^
  File "D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system\venv\Lib\site-packages\langchain_core\language_models\chat_models.py", line 1297, in _generate_with_cache
    result = self._generate(
        messages, stop=stop, run_manager=run_manager, **kwargs
    )
  File "D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system\venv\Lib\site-packages\langchain_google_genai\chat_models.py", line 3051, in _generate
    _handle_client_error(e, request)
    ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^
  File "D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system\venv\Lib\site-packages\langchain_google_genai\chat_models.py", line 145, in _handle_client_error
    raise ChatGoogleGenerativeAIError(msg) from e
langchain_google_genai.chat_models.ChatGoogleGenerativeAIError: Error calling model 'gemini-1.5-flash' (NOT_FOUND): 404 NOT_FOUND. {'error': {'code': 404, 'message': 'models/gemini-1.5-flash is not found for API version v1beta, or is not supported for generateContent. Call ListModels to see the list of available models and their supported methods.', 'status': 'NOT_FOUND'}}

(venv) D:\one drive\Desktop\Notes\Work Files for Internship\Gen_AI\ASSIGN_3\resume-screening-system>
```

Figure 3: Evidence of debugging an initial connection error during development.

➢ Detailed Trace Analysis:

- The following screenshot shows the granular details of a successful execution. It displays the prompt sent to the LLM, the raw response, and the final parsed JSON output.

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}
```

Figure 4: Detailed LangSmith trace showing the modular execution of the LCEL chain.

6. Conclusion

This project demonstrates a scalable approach to HR automation. By combining LangChain's modularity with LangSmith's observability, I built a system that is not only functional but also easy to debug and audit. This project serves as a foundation for building complex, reliable LLM-based applications.